Anomaly Detection in Synthetic p99 Latency Data

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July 2, 2026

Abstract

This project investigates anomaly detection methods for synthetic p99 latency data representing distributed system performance. High-percentile latency metrics are naturally noisy, especially in low-volume environments, and anomalies may appear at different levels of the system hierarchy. The synthetic dataset models three environment types with different baseline latency and traffic volume, and five anomaly scenarios: short-term regressions, subtle local regressions, long-term degradation, and a mixed realistic scenario.

Short-term and long-term anomaly detection were evaluated separately. For short-term anomalies, rolling z-score, robust z-score, IQR, CUSUM, Kalman residual detection, Shewhart control chart, and EWMA control chart were tested. CUSUM achieved the best point-level performance, while several threshold-based methods achieved the highest event-level recall. For long-term anomalies, linear slope, Mann-Kendall, and rolling slope were evaluated, along with long-window adaptations of CUSUM, EWMA, and IQR to test whether short-term-successful methods transfer to gradual degradation. Long-term results were generally weaker: linear slope performed best overall, and among the adapted methods, long CUSUM was the most promising but still did not outperform linear slope.

1 Introduction

Performance regressions in distributed systems may affect the whole infrastructure or only a small part of the hierarchy (e.g., a single execution unit), which makes detection difficult — especially for p99 latency, a metric that is naturally noisy and sensitive to low sample sizes.

The project addresses two distinct alerting needs: **short-term detection** (sudden spikes or short regressions) and **long-term detection** (subtle, persistent degradation trends). Since these problems have different characteristics, they are evaluated separately with different methods.

To keep the evaluation reproducible, synthetic datasets with known injected anomalies were generated (fixed random seed, saved as golden CSV datasets), allowing each method to be scored against ground truth using precision, recall, F1-score, and false positive rate.

2 Dataset

The dataset represents daily p99 latency measurements over 90 days per execution unit. Each row contains: date, `env_type` (X/Y/Z), hierarchy identifiers (`macro_id`, `segment_id`, `unit_id`), `daily_invocations`, `p99_latency_seconds`, and ground-truth anomaly labels (`is_injected_anomaly`, `anomaly_type`, `anomaly_scope`, `anomaly_event_id`).

The system hierarchy has 2 macro nodes, each with 2 segments of 3 units (12 units total). Units are assigned to three environment types (Table 1); lower traffic volume produces higher natural variability, so environment Z is visibly noisier than X.

Five scenarios were generated:

Environment type	Baseline p99 latency	Daily invocations	Expected variability
X	40 s	5000	low
Y	80 s	1000	medium
Z	120 s	100	high

Table 1: Baseline characteristics of the synthetic environments.

- **A – easy short regression:** clear local regression in one X-environment unit, days 15–22, +50% p99 latency.
- **B – multi-environment short anomalies:** short-term anomalies across units from different environments, testing stable vs. noisy conditions.
- **C – hidden local regression:** a subtle unit-level anomaly, visible locally but largely masked after aggregation to the macro level.
- **D – long-term trend:** gradual degradation in one unit over an extended period.
- **E – mixed realistic system:** short spikes, multi-day regressions, a noisy-environment anomaly, a segment-level regression, and a macro-level trend combined, plus non-anomalous high-jitter points in environment Z to test robustness against false positives.

3 Short-Term Anomaly Detection

Short-term detection targets sudden spikes and short multi-day regressions (scenarios A–C, E; D is excluded as it is a trend-only scenario). All methods were applied per `unit_id`, since local regressions may not be visible after aggregation. Methods are scored point-level (TP, FP, FN, precision, recall, F1, FPR against injected labels) and event-level (an event counts as detected if at least one point inside its window is flagged).

3.1 Tested Methods

- **Rolling z-score:** $z_t = (x_t - \mu_t)/\sigma_t$ using rolling mean/std; flagged if z_t exceeds a threshold.
- **Robust rolling z-score:** same idea, using rolling median and MAD instead of mean/std for outlier resistance.
- **Rolling IQR rule:** flags $x_t > Q3_t + k \cdot IQR_t$ from the rolling window; distribution-free.
- **CUSUM:** accumulates standardized positive deviations from a rolling local baseline, well suited to sustained multi-day regressions rather than single-point spikes. [1]
- **Kalman filter:** a 1-D Kalman filter estimates a dynamic baseline; flags large positive residuals. [6]
- **Shewhart control chart:** flags $x_t > \mu_t + k\sigma_t$ from a rolling window (classical SPC). [1]
- **EWMA control chart:** exponentially weighted moving average as a recency-weighted dynamic baseline; flags large positive residuals. [5]

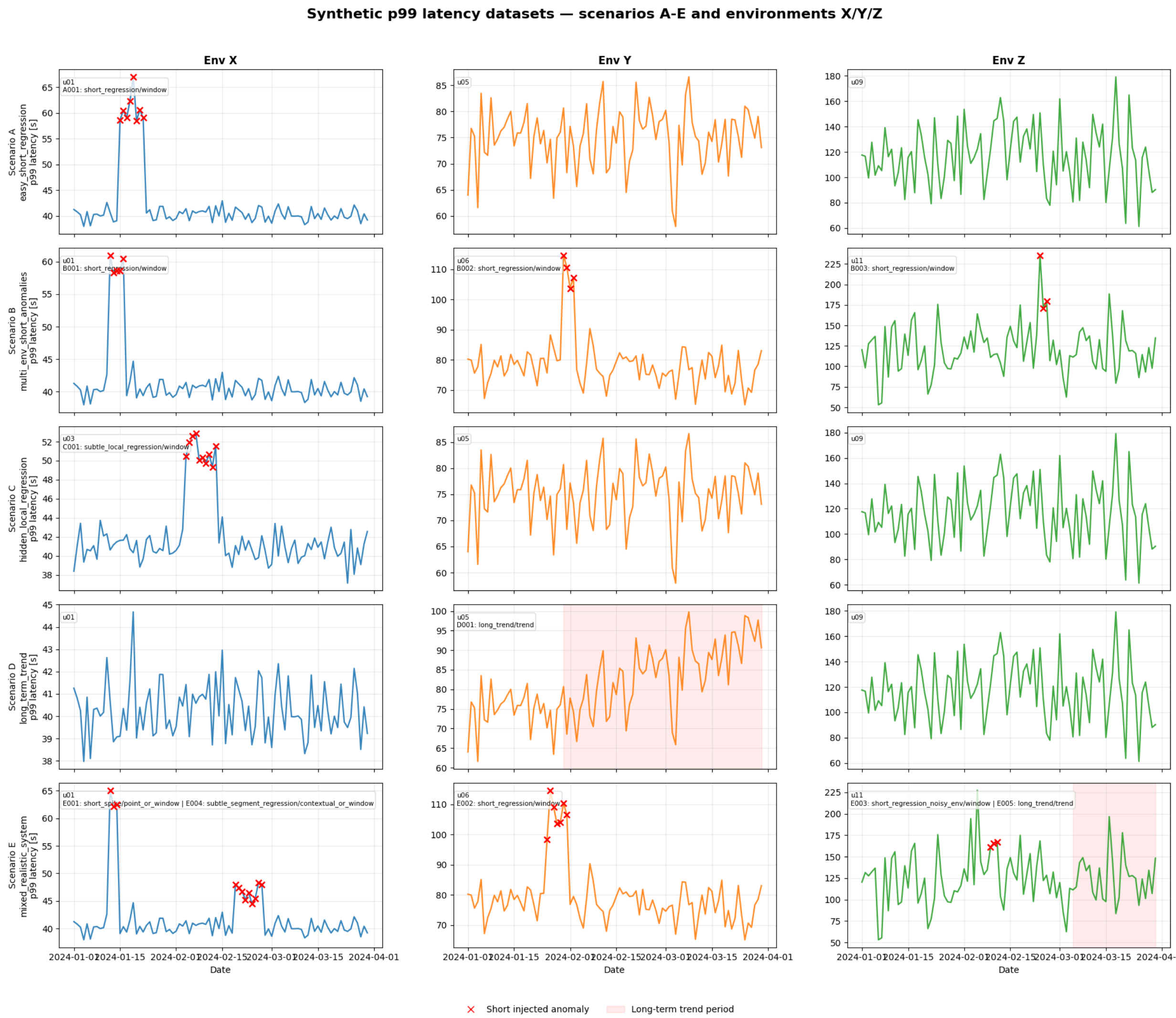

Figure 1: Overview of the generated synthetic p99 latency datasets.

3.2 Parameter Search and Best Settings

A grid search was run per method across scenarios A, B, C, E and multiple random seeds (windows 7–21 days, min. history 5–10, thresholds/multipliers as appropriate for each method; CUSUM additionally tuned over drift $k \in \{0.25, ..., 1.0\}$ and alarm threshold $h \in \{4, ..., 12\}$; Kalman over process/measurement variance). Best configurations, selected by mean F1 with FPR as a tie-breaker, are shown in Table 2.

Method	Best parameters	F1	Comment
Rolling z-score	window=21, min. hist.=10, thr.=2.5	0.326	weakest baseline
Robust z-score	window=14, min. hist.=10, thr.=3.0	0.409	more FPs than z-score
Rolling IQR	window=21, min. hist.=10, mult.=2.0	0.560	best rolling-window baseline
CUSUM	window=14, min. hist.=10, k=1.0, h=6.0	0.697	best overall short-term detector
Kalman residual	proc. var.=0.005, meas. var.=5.0, thr.=2.5, warmup=10	0.448	moderate, conservative
Shewhart	window=21, min. hist.=10, k=3.0	0.344	low FPR, low recall
EWMA	α=0.1, thr.=2.5, warmup=10	0.380	stable but misses anomalies

Table 2: Best parameter settings for short-term anomaly detection methods.

3.3 Point-Level Results

Key findings:

- Among rolling-window methods, **IQR** had the best and most stable F1 across all scenarios (fits the "temporary increase above local range" shape of the injected anomalies); standard rolling z-score was the weakest, especially in noisy environment Z.
- Among the four additional detectors, **CUSUM** was the best point-level method overall (highest F1 and recall, FPR $<$ 1%), consistent with its design for accumulating evidence over sustained multi-day regressions (Table 3).
- Performance depends strongly on environment: in X and Y, CUSUM wins on recall; in Z, all methods are either low-recall or low-precision (Table 4) because natural jitter is hard to distinguish from injected regressions.

3.4 Event-Level Results

At the event level (an anomaly counts as detected if at least one point in its window is flagged), the ranking changes: **Kalman residual, rolling z-score, EWMA and Shewhart detected the most events** (41/45 globally, recall 0.911), ahead of IQR (40/45) and CUSUM (35/45) — see Table 5. CUSUM, despite the lower event recall, had the highest mean number of detected points per event (5.84), meaning it captures more of an anomaly *once* it fires. Results for Scenario E alone follow the same ranking (best: Kalman/z-score/EWMA/Shewhart at 18/20; worst: robust

Method	Precision	Recall	F1	FPR	TP	FP	FN
CUSUM	0.778	0.490	0.734	0.0119	136	59	64
IQR	0.517	0.409	0.548	0.0082	97	43	103
Kalman residual	0.472	0.333	0.422	0.0093	80	49	120
Rolling z-score	0.417	0.316	0.386	0.0128	69	67	131
Robust z-score	0.417	0.355	0.375	0.0131	68	68	132
EWMA	0.466	0.272	0.360	0.0082	56	43	144
Shewhart	0.537	0.235	0.357	0.0052	49	27	151

Table 3: Point-level results for Scenario E, aggregated across environments and seeds.

Env.	Method	Precision	Recall	F1	FPR	TP	FP	FN
X	CUSUM	0.607	0.932	0.716	0.0202	226	139	39
X	IQR	0.695	0.630	0.638	0.0083	149	58	116
X	Kalman	0.712	0.442	0.529	0.0050	115	35	150
Y	CUSUM	0.811	0.654	0.725	0.0014	36	10	19
Y	IQR	0.338	0.750	0.653	0.0076	39	54	16
Y	Robust z-score	0.224	0.693	0.536	0.0115	35	82	20
Z	CUSUM	0.500	0.033	0.500	0.0001	1	1	29
Z	IQR	0.120	0.233	0.410	0.0053	7	38	23
Z	Kalman	0.110	0.233	0.375	0.0070	7	50	23

Table 4: Top three point-level short-term methods by environment (scenarios A, B, C, E; all seeds).

z-score at 14/20). Short regressions injected into environment Z were the hardest to detect for every method, since Z's high natural variability (Table 1) makes a genuine regression difficult to distinguish from ordinary jitter.

This shows point-level and event-level evaluation answer different questions: CUSUM is best for identifying anomalous points and estimating regression duration, while Kalman/z-score/EWMA/Shewhart are better suited to simply flagging *that* an incident occurred.

Method	Detected events	Total events	Mean detected points	Event recall
Kalman residual	41	45	3.33	0.911
Rolling z-score	41	45	2.76	0.911
EWMA	41	45	2.49	0.911
Shewhart	41	45	2.09	0.911
Rolling IQR	40	45	4.33	0.889
Robust z-score	37	45	3.64	0.822
CUSUM	35	45	5.84	0.778

Table 5: Global event-level results for short-term anomaly detection.

3.5 Low-Sample Environment Considerations

Environment Z's low invocation count drives high natural p99 variability, and every method performs worse there than in X or Y. Practical mitigations: widen the monitoring window (longer rolling windows / multi-day aggregation, at the cost of detection latency); monitor a lower percentile (p90/p95) when sample size cannot support a reliable p99 estimate; or use environment-specific, more conservative thresholds for low-volume units.

4 Long-Term Anomaly Detection

Long-term detection targets gradual p99 degradation that never produces an individually extreme daily value (scenarios D, clean trend; E, trend mixed with short-term anomalies and noise).

4.1 Tested Methods

- **Linear slope:** fits a linear regression per unit; flags a positive slope above threshold that is statistically significant (regression p-value). [4]
- **Mann-Kendall:** non-parametric monotonic-trend test; flags a significant increasing trend. [2, 3]
- **Rolling slope:** local linear trend inside a moving window, more sensitive to trends confined to part of the series or partially masked by noise.

Three additional methods, adapted from the short-term detectors, were also tested to see whether short-term-successful approaches transfer to gradual degradation:

- **Long CUSUM:** same accumulation logic as short-term CUSUM, with a longer baseline window and lower drift sensitivity.
- **Long EWMA:** smoothed dynamic baseline intended to reduce daily p99 jitter.
- **Persistent IQR:** rolling-IQR threshold that must be exceeded for several consecutive days, to avoid treating isolated spikes as trends.

Grid search (min. points 20–45, slope threshold 0.05–0.20, p-value 0.01–0.10, rolling window 21–45 days) selected: linear slope (min. pts=20, slope thr.=0.05, p=0.05); Mann-Kendall (min. pts=20, p=0.05); rolling slope (window=30, min. periods=30, slope thr.=0.10) — chosen by F1 with FPR as tie-breaker.

4.2 Results

Key findings:

- In Scenario D (clean monotonic trend), linear slope and Mann-Kendall detected every true anomaly point (recall 1.000, F1 0.808) — global trend tests work well when degradation is consistent.
- Rolling slope had reasonable recall but many more false positives, likely from temporary upward moves unrelated to the injected trend.
- Long CUSUM was the best of the adapted methods but still weaker than the trend tests.
- In Scenario E, where the trend is partially hidden by short-term anomalies and noise, the ranking flips: rolling slope achieves the best F1 and recall, while the global trend tests and long CUSUM degrade sharply.

Scenario	Method	TP	FP	FN	Precision	Recall	F1	FPR
D	Linear slope	305	145	0	0.678	1.000	0.808	0.028
D	Mann-Kendall	305	145	0	0.678	1.000	0.808	0.028
D	Rolling slope	251	620	54	0.288	0.823	0.427	0.122
D	Long CUSUM	239	189	66	0.558	0.784	0.652	0.037
D	Long EWMA	76	157	229	0.326	0.249	0.283	0.031
D	Long IQR	35	12	270	0.745	0.115	0.199	0.002
E	Rolling slope	420	703	360	0.374	0.538	0.441	0.152
E	Linear slope	260	640	520	0.289	0.333	0.310	0.139
E	Mann-Kendall	260	640	520	0.289	0.333	0.310	0.139
E	Long CUSUM	226	525	554	0.301	0.290	0.295	0.114
E	Long EWMA	72	199	708	0.266	0.092	0.137	0.043
E	Long IQR	9	115	771	0.073	0.012	0.020	0.025

Table 6: Long-term anomaly detection results by scenario, best parameter configuration per method.

Method	TP	FP	FN	Precision	Recall	F1	FPR
Linear slope	565	785	520	0.419	0.521	0.464	0.080
Mann-Kendall	565	785	520	0.419	0.521	0.464	0.080
Rolling slope	671	1323	414	0.337	0.618	0.436	0.136
Long CUSUM	465	714	620	0.394	0.429	0.411	0.073
Long EWMA	148	356	937	0.294	0.136	0.186	0.037
Long IQR	44	127	1041	0.257	0.041	0.070	0.013

Table 7: Overall long-term anomaly detection results, aggregated across Scenarios D and E.

- Aggregated across both scenarios (Table 7), linear slope and Mann-Kendall have the best F1 overall (driven by Scenario D), rolling slope has the highest recall at the cost of FPR, and long EWMA/IQR perform poorly — threshold-exceedance approaches do not transfer well to gradual trends even with longer windows and consecutive-day requirements.

5 Summary

Anomaly detection difficulty depends strongly on both anomaly type and environment stability. For **short-term** anomalies, CUSUM gives the best point-level performance for sustained multi-day regressions, but event-level evaluation favors simpler threshold methods (Kalman, rolling z-score, EWMA, Shewhart) when the goal is only to trigger an alert during an incident — so the best choice depends on whether the system needs to characterize the regression or just detect that one occurred.

For **long-term** anomalies, results are weaker overall: linear slope performs best, Mann-Kendall/rolling slope/long CUSUM are useful but less consistent, and long EWMA/IQR perform poorly by missing most trend points — gradual degradation is harder to detect reliably than short-term spikes.

Low-volume environments (Z) are consistently the hardest to monitor, since high natural variance masks injected anomalies. Practical monitoring systems should consider wider aggregation windows, environment-specific thresholds, or lower latency percentiles for such units.

References

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